



# Web and Mobile Ticketing and Service Tracking for Customer Engagement of a Small Accounting Firm

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## ABSTRACT

The abundance of free technology tools that can be used to communicate of today's businesses are in way creating more confusion and redundant tasks which is not good for a striving small business. The study aims to serve as a single platform of communication tool for a small accounting firm, a website and mobile application were developed which are the capable of providing an online ticketing which can respond to customer concerns, a service tracking, and a tool that can aggregate reports of the firm's performance in providing services in the form of charts. Web Development Life Cycle (WDLC) was used as the process model in the development of the website and mobile application. Interview validated the business processes and supported the design of web and mobile applications. Development tools used for website and mobile application are Cascading Style Sheets (CSS), Hypertext Preprocessor (PHP), Android and MySQL for database. In order to evaluate the performance of the developed system, a usability evaluation was conducted with the customers and employees of the accounting firm using the criteria: communication, transaction, aesthetics, ease of use, information and maintainability. Evaluation result shows that majority of customers strongly agree on the efficiency of communication and transaction functionalities of the developed system with the need in the improvement of image and text. Meanwhile, the evaluation of website's back-end shows promising result as well with mostly strongly agreement except for the transaction criterion wherein the employees who used the developed system had difficulty in using the system. Overall, the visual components, management of tickets, management of service tracking and viewing of the aggregate reports in the form of charts are all beneficial to a small business so that they could

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focus on their core business and use the developed website and application in customer engagement.

## CCS Concepts

• **Applied computing** → **Enterprise computing** → **Business process management** → **Business process monitoring**

## Keywords

Website; Mobile Application; Customer Engagement; Service Tracking; Online Ticketing; Communication Tool.

## 1. INTRODUCTION

Today, technologies have proven to be effective in many applications, not only for its functionality but also with its effectiveness and purpose, particularly in business like Micro, Small, Medium Scale Enterprises (MSMEs). MSMEs is defined as any business activity/enterprise engaged in industry, agribusiness/services, whether single proprietorship, cooperative, partnership, or corporation whose total assets, inclusive of those arising from loans but exclusive of the land on which the particular business entity's office, plant and equipment are situated and MSMEs can be based on the number of employees and asset size.[1] MSMEs play a major economic role in the country because of employment opportunities and rural industrialization.[2] In the Philippines, the growth and development of MSMEs increases as they use resources more efficiently than large enterprises.

Information Technology opened doors to business owners and other users that enable the firm to provide the customers with efficient communication. Today's customers have higher expectations than before and services are provided to them through various communication channels such as e-mail, website and mobile technologies.[3]

The use of emerging technologies such as website and mobile technologies has an impact into modern businesses as it brought tremendous changes to the marketing activities, tools and techniques including managing of customer relations and improving long-term customer relationships to enhance their individual potential.[4]

And in this era of technology, CRM that consists of e-CRM (Electronic Customer Relationship Management) is acknowledged

as another potential solution for business.[5] E-CRM can bring great benefits to the firm as it can increase productivity, improve customer relationships, and keep the information transparent. The emergence of these technologies improve the traditional way of communicating and providing customer service [6] making the firm to be more accessible and providing convenience on both the firm and customers. Also, the use of communication tools in the business can serve as the medium for customer engagement and interaction. Highly engaged customers can increase customer loyalty. Higher customer satisfaction can be achieved through offering better services and improving communication. Communication is the source of all collaboration and relationships.[7]

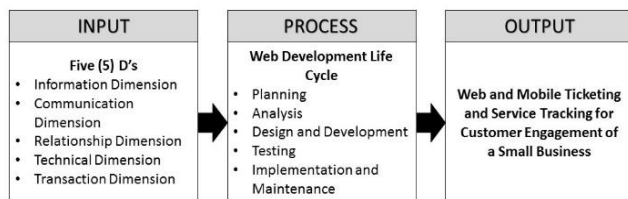
Digital technologies have been innovating customer service and service tracking in business organizations. Customer engagement is the result of the interactions between customers and organization. Interaction is defined as “any place at which a company seeks to manage a relationship with a customer, whether through people, technology, or some combination of both”.[8] It is also “the process of strategically managing a customer’s entire experience with a product or company”.[9]

A typical MSMEs has been missing opportunities in interacting with their customers because of using different means of communication such as email, instant messaging and social media sites which may also increase the workload of the employees just by organizing and integrating the collected data from inquiries and other transactions. Thus, this study aims to help a small accounting firm to respond to the customers need promptly through the development of communication tool via web and mobile application.

The tool is to be used primarily to provide status of the services availed by customers and a venue for customers to communicate their concerns and feedback. Also, the tool includes charts showing the performance of the accounting firm in providing services to their customers. In particular, a website for online ticketing was developed which aims to eliminate the redundancy that typically comes from lack of communication and allows the workforce to be used more effectively in understanding the customer needs [10] and a mobile application that can be used by customers to track the progress and status of the availed services. Accordingly, creating modern communication tools such as website and mobile application allow the firm to be more reachable which may promote good relationship and engagement with their customer.

## 2. METHODS

A basic input-process-output was used for the research framework of the study as illustrated in Figure 1.



**Figure 1** Figure 1. Research Framework of the Study

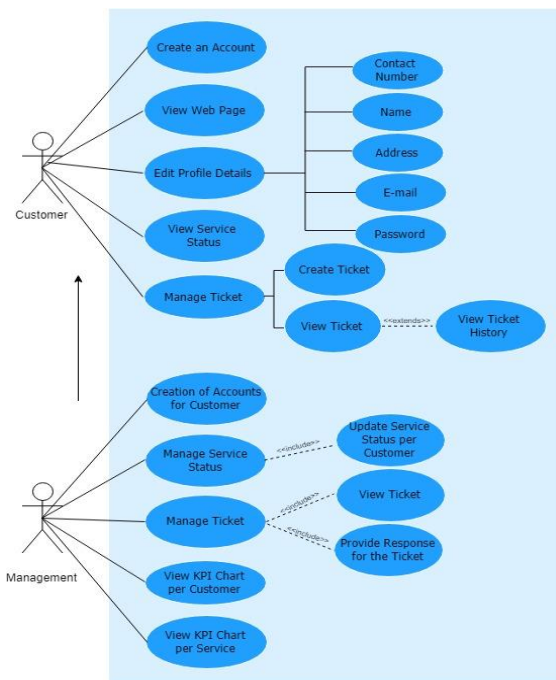
**Input.** In this phase, these five (5) dimensions are used as input in order to provide a web and mobile ticketing and service tracking for customer engagement: (1) information dimension are the updated and accurate information in the website and mobile application which consists of services offered, news and events, photo gallery, and location/direction map for marketing and promotion of the business; (2) communication dimension is the capability of the visitors to make feedbacks through the search capabilities and contact information of the business; (3) relationship dimension are the policies and rules that the implements for the services they are providing for clients; (4) technical merit dimension are the tools used in developing the website and mobile application such as Cascading Style Sheets (CSS), Hypertext Preprocessor (PHP), Android and MySQL for the database; and, (5) transaction dimension is the tracking of services availed.

**Process.** Web Development Life Cycle (WDLC) was used to process the inputs of the system. Web Development Life Cycle would be the process on how the website will be created and developed. It followed the series of steps indicated in the WDLC. The order of progression would be: planning, analyzing, design and development, testing, and implementation and maintenance.[11]

- **Planning.** In this phase, data gathering was conducted through interview and survey in order to get the essential information for the development of the system. Functions and business processes were also observed. The website and mobile application were identified to be used as a marketing tool for the business in order to advertise the different services they offer, to provide their online customer concerns and to track the services that the customers availed. The target audience for the website and mobile are the customers interested in the field of accounting, auditing and taxation.
- **Analyzing.** During this phase, the data collected in the planning phase was used to develop the functionalities and content of the website and mobile application. The decisions about the website content and functionality were identified.
- **Design and Development.** In this phase, the contents and the scope of the website are defined and organized. The established standards in the implementation of the services it provides helped a lot in the design and development of the system. Some of the established business processes that were part of the system includes: business registration, bookkeeping, tax and other statutory compliance, advisory services, and audit and assurance service. Figure 2 shows the functionalities or the things that can be performed by the users of the system.

The website’s front-end includes home page, about us page, our services, contact us, gallery, and when a customer is logged to the system, the customer can edit profile, view service status, and manage ticket status such as creating ticket and viewing of ticket status. The website’s back-end includes the management of the website for adding of new account, managing tickets of customers, viewing of KPI’s chart per customer and per service. The functionality and features of the mobile application for customer includes editing of profile, viewing and managing of service status and creation of ticket when they have some inquiries or problems encountered during and post service.

- **Testing.** During this phase, the website contents, functionality and usability are tested. Users including the employees of the firm were tasked to perform the features in order to measure the ease of use of the website and mobile application. Testing of the functionality and content involves ensuring that links are linked correctly, page elements are interactive and testing the website into different browser and version to check its compatibility. KPI that is incorporated on service tracking were also tested in order to determine if the KPI is working accurately according to the specific service. In addition, usability testing was used to measure the user's perception of the experience on the website. The usability testing was focused on the main aspects including suitability, efficiency and usability. Each aspect was tested through observation in thorough navigation on the website.



**Figure 2. Use Case Diagram of the System**

- **Implementation and Maintenance.** After usability testing was completed, the website and mobile app are ready for implementation and maintenance. The developers uploaded the website into a web server. After the website has been uploaded, maintenance of the website was done by updating the content through removing out-dated information and replacing a new one.

**Output.** The final outcome consists of the following tools: web and mobile ticketing and service tracking for customer engagement of the accounting firm. The tools are capable of monitoring and tracking the services they have been providing such as bookkeeping, auditing, tax consultancy, advisory, assurance services, business registration and accounting works. In addition, since the firm values urgency and efficiency in the performance of its services, thus, the Key Performance Indicator (KPI) of the firm was integrated in the service tracking to evaluate

the success of their activities and engagements which is set at least a day before the government's deadline.

### 3. RESULTS AND DISCUSSION

Usability Testing is a phase in software testing process wherein the software was tested by the actual respondents to evaluate the usability of the prototype. In this study, there are 28 front-end respondents and two back-end respondents. Front-end respondents are the existing customers of the accounting firm while the back-end respondents are the clerk and manager of the accounting firm.

#### 3.1 Usability Evaluation of Front-End

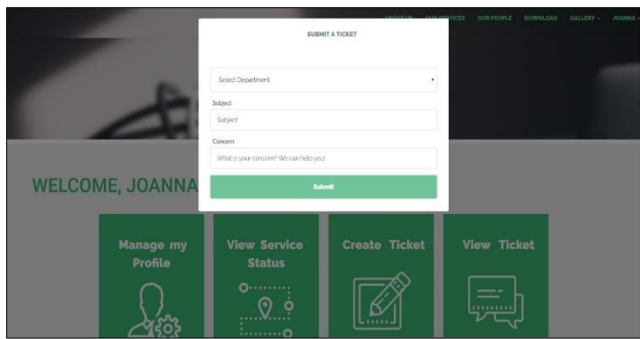
The usability testing questionnaire for the usability evaluation of the website's front-end was patterned after the Five (5) Dimension Framework namely: (1) communication, (2) transaction, (3) visual aesthetics, (4) ease of use, and (5) information. The criteria used for the survey aims to get the perception of the respondents or users for further improvement of the prototype and alignment of the business processes with the system. There were 28 customers that were able to use the website and mobile application and were asked to answer a 4-point scale questionnaire consisting of 22 statements regarding their perception on the use of the system.

Communication criterion aims to answer the question on providing an online ticketing that can respond to customer concerns and questions promptly. Transaction criterion aims to answer the question whether the firm has provided a tracker that can allow the customers to know the status of the services availed and management of accounts. Visual aesthetics aims to identify whether the design is appropriate and has pleasing appearance. Ease of Use aims to identify whether the prototype is user-friendly and easy to navigate. Lastly, Information category aims to identify whether the information is accurate and up-to-date. Table 1 shows the summary of survey results.

**Table 1. Summary Results of the Front-End's Website and Mobile Application**

| Criteria             | Strongly Agree | Agree | Disagree | Strongly Disagree |
|----------------------|----------------|-------|----------|-------------------|
| 1. Communication     | 94.5%          | 5.5%  | 0.0%     | 0.0%              |
| 2. Transaction       | 94.5%          | 5.5%  | 0.0%     | 0.0%              |
| 3. Visual Aesthetics | 87.7%          | 12.3% | 0.0%     | 0.6%              |
| 4. Ease of Use       | 88.5%          | 11.5% | 0.0%     | 0.0%              |
| 5. Information       | 91.2%          | 8.8%  | 0.0%     | 0.0%              |
| Overall              | 91.3%          | 8.7%  | 0.0%     | 0.1%              |

In communication, the criterion shows that majority of the users strongly agree of the efficiency brought by the system. The statements under this criterion deal on how the respondents are able to perform the features on the ticketing such as filing and receiving a ticket. Figure 3 shows the submission of ticket page wherein the user needs to provide details of their concern and will be directed to the department that could help them solved their problems.



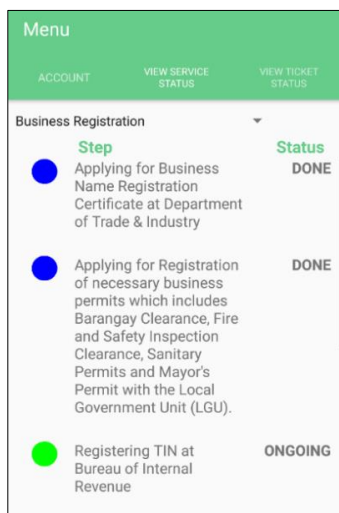
**Figure 3. Submission of Ticket Page for Clients**

The statements on transaction criterion deal with tracking of the services availed and updating of account. Result shows that almost all of respondents were able to update their account and were able to easily know the current status of the service they availed. Figure 4 shows the status service page for customers.

| ID | SERVICE                       | Start                  | End                    | STATUS    |
|----|-------------------------------|------------------------|------------------------|-----------|
| 21 | Business Registration         | Mar. 15, 2018 12:16 AM | Apr. 11, 2018 12:16 AM | Completed |
| 43 | Business Registration         | Mar. 15, 2018 12:16 AM | Apr. 11, 2018 12:16 AM | Completed |
| 46 | Business Registration         | Mar. 25, 2018 1:35 PM  | Apr. 21, 2018 1:35 PM  | Active    |
| 10 | Business Registration Renewal | Mar. 15, 2018 12:21 AM | Apr. 10, 2018 12:21 AM | Completed |
| 10 | Bookkeeping                   | Mar. 15, 2018 12:24 AM | Mar. 20, 2018 12:24 AM | Completed |
| 22 | Bookkeeping                   | Mar. 25, 2018 2:03 PM  | Mar. 30, 2018 2:03 PM  | Active    |
| 5  | Audit                         | Mar. 15, 2018 12:52 AM | May. 19, 2018 12:52 AM | Completed |
| 3  | Agree-upon                    | Mar. 15, 2018 12:54 AM | Apr. 14, 2018 12:54 AM | Completed |
| 13 | Tax                           | Mar. 5, 2018 4:00 PM   | Mar. 10, 2018 4:00 PM  | Completed |

**Figure 4. Service Status Page for Customer**

In the service status page, the customer would be able to view history of the services they have availed from the firm. Also, the customer can view the details per service and they can review it if the process they have agreed on are being followed as shown in the mobile application of Figure 5. Both the website and mobile application would show the same details.



**Figure 5. Service Status in Mobile App for Customer**

In visual aesthetics, the result shows that the font style used is readable; however, the respondents do not quite agree with the text display thus making this criterion the lowest. Overall, the visual aesthetics of the front-end scored fairly in terms of concept,

elements such as clarity of images and design of the website and mobile application.

In terms of ease of use, the respondents agree on the ease of use of both the website and mobile application. The result shows that majority of the respondents strongly agree that the website and mobile app are user-friendly and some respondents commented that the quick links are helpful. Respondents claimed that they finished the transaction in a short amount of time due to the user-friendliness of both the website and mobile application.

Lastly, the statements in the information criterion deal with how accurate and useful the information are on the website and mobile application. The respondents strongly agreed that the information on the website and mobile application are accurate and useful.

### 3.2 Usability Evaluation of Back-End

Similarly, the back-end's evaluation was based on the Five Dimension Frameworks: (1) communication, (2) transaction, (3) visual aesthetics, (4) ease of use, and (5) maintainable platform. The criteria for usability evaluation of the back-end of the website are similar with the front-end except for information which is replaced by maintainable platform. Also, the respondents for this survey are the employees of the firm who will be given the task of performing the maintenance of the system such as checking of the service status and other inquiries or concerns of the customers. Maintainable platform aims to answer the question whether the system is capable of aggregating reports through graphs and whether the back-end respondents were able to manage information for the website. Table 2 shows the summary results on the survey of the back-end's website.

**Table 2. Summary Results of the Back-End's Website**

| Criteria                 | Strongly Agree | Agree        | Disagree    | Strongly Disagree |
|--------------------------|----------------|--------------|-------------|-------------------|
| 1. Communication         | 83.3%          | 16.7%        | 0.0%        | 0.0%              |
| 2. Transaction           | 75.0%          | 25.0%        | 0.0%        | 0.0%              |
| 3. Visual Aesthetics     | 85.7%          | 14.3%        | 0.0%        | 0.6%              |
| 4. Ease of Use           | 87.5%          | 12.5%        | 0.0%        | 0.0%              |
| 5. Maintainable Platform | 87.5%          | 12.5%        | 0.0%        | 0.0%              |
| <b>Overall</b>           | <b>83.8%</b>   | <b>16.2%</b> | <b>0.0%</b> | <b>0.1%</b>       |

In communication criterion, the result shows that majority of the respondents were able to receive ticket from customers, have received notification that the ticket filed by the customer has been successfully transmitted, and, able to respond to the ticket filed by the customer as well. Figure 6 shows the viewing of tickets submitted by the customers in the website. It shows the list of customers and their tickets and its status, last update, and action taken. To display the details of the ticket such as the subject, time details of the sent ticket and the name of the sender as well as the name of the responder, the user should simply need to click on a particular customer.

| VIEW TICKETS                |                               |         |                              |                |        |
|-----------------------------|-------------------------------|---------|------------------------------|----------------|--------|
| Show 25 entries             |                               |         | Search: <input type="text"/> |                |        |
| Customer                    | Service                       | Subject | Status                       | Last Update    | Action |
| joanna                      | Agreed-Upon Procedures        | hello   | On-going                     | April 25, 2018 | Delete |
| joanna                      | Bookkeeping                   | new     | On-going                     | April 25, 2018 | Delete |
| joanna                      | Business Registration Renewal | new ult | On-going                     | April 25, 2018 | Delete |
| joanna                      | Business Registration         | test    | Done                         | April 5, 2018  | Delete |
| Showing 1 to 4 of 4 entries |                               |         |                              |                |        |
|                             |                               |         | Previous                     | 1              | Next   |

**Figure 6. View Tickets Page (Back-end)**

The statements on transaction criterion deal with tracking of the services and creating of account. The result shows that it is the lowest percentage among the criterion. According to the respondents, they were not quite successful in managing the service availed by the customer since it is their first time to have used it; however, they are certain that one they are familiar with it, it would be easier for them to manage the customer's account.

In the visual aesthetics criterion, the result shows that majority of the respondents strongly agree that the elements and design of the website back-end is pleasing and text descriptions are easy to understand; however, the text display and images has to be improved. The management of the accounting firm complimented the design elements used in service tracking.

The result of the ease of use criterion shows that the website's back-end is easy to navigate and user friendly except that it took them a little while in finishing the back-end transactions.

Lastly, in maintainable platform, the result shows that all the respondents were able to view the KPI charts per customer and per service; however, there were difficulty in managing the web pages such as updating and deleting necessary information. Figure 6 displays the view KPI page of a customer on the website's back-end.



**Figure 6. View KPI Page per Customer**

## 4. CONCLUSIONS

The size of the company should not hinder them in customer engagement with the use of technological tools; however, small businesses had the concept of being not ready or not needing it. This study aims to prove that technology can help even small businesses improve their customer engagement. The usability evaluation of the front-end of the web and mobile application shows that customers of the accounting firm are ready in terms of technology and skills. Majority of the customers perceived that the communication capability of the tools can be used to promote engagement and it provides the information they needed with less intervention from the accounting firm. Customers can now communicate with the accounting firm's management regarding any concerns through the online ticketing which can be used by

the management to gain insights about the customer's concerns, inquiries and opinions. In terms of platform maintainability, the back-end users of the application had a difficulty; however, with continuous use and training the back-end users are expected to be familiar with its usage. Finally, the performance of the accounting firm in terms of providing services can be easily aggregated through the web and mobile application. The management can now gain insights on the performance of their services in the form of charts for ease of interpretation.

## 5. RECOMMENDATIONS

Majority of the respondents complimented the website and mobile application for having service tracking and online ticketing; however, the developed system for the accounting firm still has to be improved when used in the other branches that they are planning to put up. The study recommends integrating urgency or priority level on ticketing to immediately attend to inquiries or concerns of the customers. Also, the study recommends providing a feature on ticketing which can send an attachment such as file, image or video to enable the users to send or transfer relevant files directly so that the users can easily share more information about their issues. Lastly, since the developed system is for android users, it is recommended that iOS version of the mobile application can be developed for iOS users.

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